

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.05, beta.var.scale = 0.3\n')
```

```
## sdev = 0.05, beta.var.scale = 0.3
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.2194  0.2450  0.2522  0.2527  0.2594  0.2850

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.0950  0.1389  0.1352  0.1700  0.9239
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.00000 0.09778 0.14222 0.13806 0.17333 0.73778
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.000000 0.005000 0.008333 0.102147 0.031111 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.078190 0.024028 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.006111 0.015000 0.162432 0.117778 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04000 0.05667 0.06111 0.06099 0.06556 0.07944

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05556 0.10167 0.10544 0.14056 0.89000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05778 0.10556 0.10824 0.14444 0.69111
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001667 0.060000 0.415793 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0055556 0.2493771 0.3769444 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1289 1.0000 0.6985 1.0000 1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.009444 0.021111 0.024444 0.024824 0.028333 0.044444

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02778 0.07222 0.07851 0.11111 0.87278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02889 0.07611 0.08171 0.11667 0.60889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.9378  1.0000  0.7980  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0005556 0.2262954 0.1736111 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.06722 1.00000 0.67911 1.00000 1.00000

--- Summary for K = 4 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.002222 0.011111 0.013889 0.014650 0.017222 0.058889

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02347 0.06000 0.06964 0.09778 0.83111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06444	0.07317	0.10389	0.66056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5644	0.8789	0.7161	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2748	0.6607	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8701	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.007778	0.010000	0.010938	0.012778	0.061111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04500	0.05667	0.08111	0.81722

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.04889	0.06039	0.08778	0.70000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3339	0.8789	0.6657	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2825  0.8436  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000  1.0000  0.8846  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.0005556 0.0044444 0.0061111 0.0066622 0.0077778 0.0461111

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01222 0.03722 0.04926 0.06889 0.73722
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01444 0.04111 0.05321 0.07556 0.55000
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.6900  0.5395  1.0000  1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.9467 1.0000  1.0000  1.0000  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2434  0.4363  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8344  1.0000  0.8041  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.001667 0.005556 0.007778 0.007989 0.010000 0.017778

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

0.000000 0.005556 0.025000 0.039213 0.056111 0.751111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.028889	0.042689	0.062778	0.480000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3453	0.9767	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9984	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1355	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.930000	0.641091	1.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.004444	0.006111	0.006172	0.007778	0.016667

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.026667	0.041085	0.057778	0.753333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.007778 0.031111 0.045411 0.065556 0.494444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.3621 0.9794 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 1.0000 1.0000 0.9992 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1396 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.05833 0.94000 0.65034 1.00000 1.00000
```

--- Summary for K = 9 ---

```
Z.ar:
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.002222 0.007222 0.010000 0.010361 0.012778 0.026667
```

```
beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.026667 0.043231 0.058333 0.715000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
```

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.007778 0.031111 0.047342 0.067778 0.507778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2213 0.4133 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1167 1.0000 1.0000 0.9998 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1007 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
```

```
0.0000 0.0000 0.4328 0.4537 0.9161 1.0000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2194 0.2450 0.2522 0.2527 0.2594 0.2850
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0950 0.1389 0.1352 0.1700 0.9239
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09778 0.14222 0.13806 0.17333 0.73778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.102147 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.078190 0.024028 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015000 0.162432 0.117778 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.04000 0.05667 0.06111 0.06099 0.06556 0.07944
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.05556 0.10167 0.10544 0.14056 0.89000
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.05778 0.10556 0.10824 0.14444 0.69111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001667 0.060000 0.415793 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0055556 0.2493771 0.3769444 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1289 1.0000 0.6985 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.009444 0.021111 0.024444 0.024824 0.028333 0.044444
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02778 0.07222 0.07851 0.11111 0.87278
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02889 0.07611 0.08171 0.11667 0.60889
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9378  1.0000  0.7980  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0005556 0.2262954 0.1736111 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.06722 1.00000 0.67911 1.00000 1.00000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.002222 0.011111 0.013889 0.014650 0.017222 0.058889
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02347 0.06000 0.06964 0.09778 0.83111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02556 0.06444 0.07317 0.10389 0.66056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5644  0.8789  0.7161  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2748  0.6607  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.8701  1.0000  1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.003333 0.007778 0.010000 0.010938 0.012778 0.061111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.014444 0.045000 0.05667 0.08111 0.81722
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.01556 0.04889 0.06039 0.08778 0.70000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.3339 0.8789 0.6657 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.0000 0.0000 0.2825 0.8436 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 1.0000 1.0000 0.8846 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0044444 0.0061111 0.0066622 0.0077778 0.0461111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.01222 0.03722 0.04926 0.06889 0.73722
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01444 0.04111 0.05321 0.07556 0.55000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.6900  0.5395  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9467  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2434  0.4363  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8344  1.0000  0.8041  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.005556 0.007778 0.007989 0.010000 0.017778
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.025000 0.039213 0.056111 0.751111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.028889 0.042689 0.062778 0.480000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.3453  0.9767  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9984  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1355  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003889 0.930000 0.641091 1.000000 1.000000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001111 0.004444 0.006111 0.006172 0.007778 0.016667
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.026667 0.041085 0.057778 0.753333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.031111 0.045411 0.065556 0.494444
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.3621  0.9794  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9992  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1396  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05833 0.94000 0.65034 1.00000 1.00000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.002222 0.007222 0.010000 0.010361 0.012778 0.026667
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.000000 0.006667 0.026667 0.043231 0.058333 0.715000
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.000000 0.007778 0.031111 0.047342 0.067778 0.507778
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.0000 0.0000 0.0000 0.2213 0.4133 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.1167 1.0000 1.0000 0.9998 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.0000 0.0000 0.0000 0.1007 0.0000 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.0000 0.0000 0.4328 0.4537 0.9161 1.0000
##
## ‘ ‘ ‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.